check_pg_drc

NAME

check_pg_drc
Verifies and reports technology routing design rules violations and illegal overlaps of PG net objects.

SYNTAX

```
status check_pg_drc
        [-nets {collection_of_nets}]
        [-ignore_std_cells]
        [-do_not_check_shapes_in_lib_cells]
        [-do_not_check_shapes_in_hier_blocks]
        [-ignore_clock_nets true | false]
        [-check_metal_on_track true | false]
        [-ignore_keepout_margins]
        [-check_min_metal_area_on_pins true | false]
        [-load_routing_of_all_nets]
        [-check_detail_route_shapes]
        [-coordinates {{llx lly} {urx ury}}]
        [-bottom_layer min_layer_name]
        [-top_layer max_layer_name]
        [-no_gui]
        [-output text_file]
```

Data Types

collection_of_nets	collection or list
llx	float
lly	float
urx	float
ury	float
min_layer_name	string
max_layer_name	string
text_file	string

ARGUMENTS

-nets {collection_of_nets}
Specifies the PG nets to verify. By default, the shapes of all PG nets are checked. The set of specified nets should be a subset of all PG nets. All shapes of the specified nets are validated for DRC errors against all other PG nets in the design. If there are several PG nets in the design, such as VDD1, VDD2, VSS1, VSS2, you can choose to just check some PG nets. For example, you can choose to validate only VDD2 net shapes against the other PG nets.

-ignore_std_cells
Prevents the command from loading internal objects, such as pins and internal blockages, of standard library cells. This might be useful, for example, if placement of the cells is not "legalized". By default, the command loads and checks all internal objects of such cells.

-do_not_check_shapes_in_lib_cells
Prevents the command from reporting DRC errors that involve internal shapes of the same library cell only. Cells under this option include cells of the following design_type: lib_cell, filler, end_cap, well_tap, and diode. By default, such errors are reported. For example, a small unconnected pin included in the PG net might cause "minimal metal area" error in a report.

-do_not_check_shapes_in_hier_blocks
Prevents the command from reporting DRC errors that involve internal shapes of any cell other than library cells described in this option, -do_not_check_shapes_in_lib_cells. For example, design_type macro, pad, and so on. By default, such errors are reported.

-ignore_clock_nets true | false
Specifies whether routing objects of clock nets should be ignored. By default, the tool ignores clock net objects.

-check_metal_on_track true | false
Specifies whether to check metal-shape alignment with a track on a metal layer (if corresponding rules are set in the technology file). By default, the tool does not do such checking.

-ignore_keepout_margins

Prevents the command from checking any overlaps of routing objects with keepout-margin areas. By default, all such cases are reported as illegal-overlap errors.

-check_min_metal_area_on_pins true | false

Specifies whether the command should check metal area of unconnected pins. By default, insufficient area of such pins is reported as an error.

-load_routing_of_all_nets

The tool considers existing routing shapes and vias of all nets for any DRC errors. By default, routing shapes and vias of non-PG nets are completely ignored. That is, by default, PG shapes and vias are verified against other PG shapes and vias. This option allows the checker to check existing routing shapes and vias of all nets. The shapes that are verified still need to be of type PG and should not have shape_use detail_route.

-check_detail_route_shapes

When this option is specified, the command takes into account routing shapes and vias of PG nets which have usage attribute "detail_route" or "shield_route". By default, such shapes are ignored. This holds for shapes to be verified and the shapes to be verified against. For checking against non-PG terminals, set the option as true.

-coordinates {{llx lly} {urx ury} ...}

Specifies a rectangular area where to perform design rule checking. llx and lly are coordinates that define the lower-left corner of the area. urx and ury are coordinates that define the upper-right corner. By default, PG DRC checking is performed on the entire design.

When this option is specified, the tool considers objects within the specified rectangle, and nearby objects (that are within the range of DRC checker) associated with the PG nets.

-bottom_layer min_layer_name

Specifies the bottom routing layer on which to perform DRC checking. If this option is specified, the tool ignores objects on layers lower than min_layer_name. By default, the tool checks down to the lowest routing layer of the design.

-top_layer max_layer_name

Specifies the top routing layer on which to perform DRC checking. If this option is specified, the tool ignores objects on layers higher than max_layer_name. By default, the tool checks up to the highest routing layer of the design.

-no_gui

Prevents the tool from writing to the GUI error set. If this option is specified, a text report is written to the file specified by the -output option. If -output is not specified, the tool writes the report to a file named "./pg_routing_errors.txt".

-output text_file

Specifies the name of the output report file. By default, the tool does not write a text report, unless the -no_gui or this option is specified. If -no_gui is specified and -output is omitted, the tool writes the report to a file named "./pg_routing_errors.txt".

DESCRIPTION

This command verifies and reports violations of technology design rules and illegal overlaps of objects (shapes, vias, and pins) of power and ground (PG) nets.

By default, only the top-level shapes of PG nets whose shape_use attribute is set to any of the following values are verified for DRC errors: follow_pin, lib_cell_pin_connect, macro_pin_connect, pg_augmentation, ring, stripe, and user_route. Checking is performed either on the entire design area, or in the area specified by the -coordinates option.

The results from this command are available as an error set for browsing with the GUI error browser, or as text report that is written to a file (when the -no_gui -output options are specified), or both.

For the text report, the command writes the error type, violated rule, layer, and other details regarding the error as follows.

```
=================
Error type:      insufficient space between two diff-net via-cuts
Violated rule:   Cut-Min_Spacing-Rule
Layer:           CO
Minimal spacing: 0.05
Actual  spacing: 0.0238(southwest - northeast)
Spacing box:     {900.621 412.336} {900.633 412.357}
Via #1:          box = {900.579 412.294} {900.621 412.336}
                      net = VDD
Via #2:          box = {900.633 412.357} {900.675 412.399}
                 net = VSS
```

The command can use multiple CPU cores to generate the report, based on the set_host_options -max_cores command setting. If this value is not set, the command runs on a single core and therefore the report generation might be slower.

EXAMPLES

The following example checks technology design rules for all PG nets in the entire design. The report is generated as an error set for the GUI error browser. The command can use up to six CPU cores:

```
prompt> set_host_options -max_cores 6
prompt> check_pg_drc
```

The following example checks only objects of the VDD net on layer MET4. The text format report is written to a file named "./net_vdd_on_M4.drc":

```
prompt> check_pg_drc -no_gui -output ./net_vdd_on_M4.drc -nets VDD \
   -bottom_layer MET4 -top_layer MET4
```

The following example checks rules within the specified rectangular area and generates both kinds of report:

```
prompt> check_pg_drc -coordinates {{287.0 362.0} {412.0 530.0}} \
   -output ./PG-errors_in_window.txt
```

SEE ALSO

set_host_options(2)
gui_open_error_data(2)
